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Psychological Adjustment to Athletic Injury

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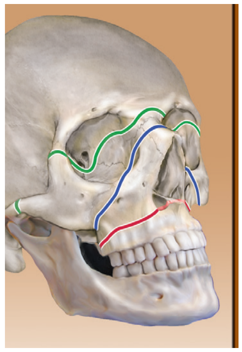

Participation in athletics involves physical risk that can result in limited ability or inability to continue sports play. Medical advances have led to the improved diagnosis of athletic injury and formulation of rehabilitative treatment plans. Psychological variables influence how individuals cope with physical injury; therefore understanding and effectively managing the psychological reaction to injury can enhance recovery from injury, facilitate return to play (RTP), and aid athletes in adapting to changing roles.

The number of persons participating in sports continues to grow. High school student participation has increased by more than 130% over the last 30 years according to the National Federation of High School Sports (NFHS), with over 7.8 million participants in 2014–2015.[1] These figures, importantly, only account for those playing scholastically, as club participation across youth sports has been estimated to be above 44 million annually.[2] Overall injury rates in high school sports were at 2.32 per 1000 athlete exposures, roughly 1.4 million for the 2015–2016 school year.[3] From the 2005–2006 school year through the 2013–2014 school year, 6.0% of the 59,862 total injuries tracked among high school athletes were career or season-ending injuries.[4] Sports-related injuries accounted for up to 1 in 5 of all emergency department (ED) visits in those under the age of 18,[5] and 19% of visits to the pediatrician's office.[6]

In a study that estimated average injury data for all 25 NCAA sports over a 5-year period from the 2009–2010 school year through the 2013–2014 school year, there were an estimated average of 210,674 total injuries per year from 28,860,299 practices and 6,472,952 competitions each year.[7] About one-fifth of these injuries required 7 days or more away from full participation in sport as part of the recovery process.[7] Among male student-athletes, football accounted for the largest number of estimated injuries per year while men's wrestling had the highest overall injury rate.[7] Among female student-athletes, soccer accounted for the largest number of estimated injuries per year while gymnastics had the highest overall injury rate.[7]

While it is common for athletes to experience athletic injury, there are many potential impacts on an athlete's emotions, thoughts, behaviors, and overall well-being. Common emotional responses include stress and anxiety,[8,9] such as fear of reinjury,[10] fear of not being able to perform at the same preinjury level, and/or to improve performance after injury,[11] sadness, depressive symptoms, loss,[12-15] isolation,[16] anger,[12,13,17] and decreased self-esteem.[13] Such negative emotions have been associated with a longer rehabilitation process,[11] decreased motivation to attend rehabilitation appointments,[12] and an increased risk of reinjury.[18]

Although many athletes are impacted in negative ways by athletic injury, each athlete may respond differently. Some athletes do not experience much disruption.[8] Some athletes are able to interpret their athletic injury as a challenge and focus on learning from the experience.[13,19] Providing support to athletes in order to increase their likelihood of experiencing these positive outcomes is critical to athlete well-being and to athletic success.

In addition, the age of the participant, the level of competition (youth, high school, club, collegiate, professional, weekend warrior), the severity of injury, and the degree of investment in the athlete identity by the participant are important factors in how the psychological response to injury is conceptualized. This chapter does not provide a comprehensive review of the developmental conceptualizations of the psychological impact of athletic injury or a comprehensive discussion of how competition level influences psychological variables. Rather, the psychology of injury is discussed more broadly, reviewing some of the general paradigms for understanding how injury psychologically impacts athletes and the literature on utilization of psychological principles to enhance recovery. Specific application of these efforts are also presented through discussion of pre-participation screening, the referral process, the value of multi-disciplinary treatment teams, and the importance of continuing to support injured athletes beyond the initial return to sport.

HISTORICAL PERSPECTIVE AND EVOLUTION

Psychological models of injury were initially divided into two categories: Stage Models and Cognitive Models.[20] Stage Models were largely adapted from examining psychological reactions to terminal illnesses. Injury is conceptualized as triggering grief and loss, because of the resulting perceived loss of an aspect of the self.[21] Commonly applied models of the stages of grief are adapted from Kubler-Ross.[22] Limitations included the reduced ability to incorporate the variety of athletes' perceptions of injury and the contextual differences that influence the psychological consequence of the injury. These limitations gave rise to Cognitive Models, which borrowed significantly from the stress and coping theory. Rather than stages, the interpretations of the injury itself are the focal point. These include cognitive attributions for injury,

Abstract

While athletic injury is a common experience for individuals who participate in sport, there are many potential impacts on an athlete's emotions, thoughts, behaviors, and overall well-being, which can also impact their physical recovery process. This chapter discusses the psychological adjustment to athletic injury topic in a broad way, reviewing the general paradigms for understanding how injury psychologically impacts athletes and the literature on utilization of psychological principles to enhance recovery. Specific applications of these efforts are also presented through discussion of pre-participation screening for potential mental health concerns. The value of specifying a referral process to a licensed mental health provider, using multi-disciplinary treatment teams, and providing ongoing support to injured athletes beyond the initial return to sport are explored. Additionally, we discuss the potential challenges in the return to sport process resulting from real or imagined pressure. Medical providers are often the staff members who interact with injured athletes soonest after injury occurs and most frequently throughout the return to sport process, so they play a critical role in recognizing potential signs of mental health concerns and referring to other members of the multi-disciplinary treatment team in order to provide holistic care for injured athletes.

Keywords

athletic injury
psychological adjustment
psychological response
biopsychosocial
integrative care

self-perceptions following injury, methods of coping, and perceived benefits of injury.[20] Subsequently, these models have been integrated to include biopsychosocial influences[23] and updated to reflect the bidirectional relationship between an individual athlete's coping tendencies and contextual factors.[18,20] See *The Handbook of Sport Psychology* for a more detailed review of this history.[24]

Some of the most widely supported models include the Stress and Injury Model, the Self-Determination Model, and the Wiese-Bjornstal Integrated Model of Psychological Response to the Sport Injury and Rehabilitation Process. The Stress and Injury Model[18] posits that heightened distress associated with the fear of reinjury saps available attention resources and peripheral vision, leading to increased muscular fatigue and reduced timing and coordination. Building off Ryan and Deci's[25] initial formulations of the Self-Determination Model, other researchers have emphasized the importance of meeting three psychological needs during the return to sport process in accordance with Self-Determination Theory: competence, or trust in one's own ability, autonomy, or a belief that one can influence their own outcomes, and relatedness, or a sense of connection to others.[26-28] The Wiese-Bjornstal Integrated Model of Psychological Response to the Sport Injury and Rehabilitation Process extends Lazarus and Folkman's 1984 cognitive appraisal theory of stress and coping and argues that preinjury variables (personality, history of stressors, coping resources, etc.), injury variables (injury history, severity of injury, etc.), and postinjury variables (cognitive appraisals made by an athlete, etc.) influence an athlete's response to athletic injury.[20,29]

PREPARTICIPATION SCREENING

Assessing for potential mental health concerns upon entry of an athlete to a club team, a collegiate program, or a professional team allows services to be provided proactively to support athletes who may be at-risk of developing mental health concerns in the future, who have already sought care in the past and wish to continue, and/or who are interested in proactively establishing support prior to experiencing stressors such as athletic injury. Some early-stage research suggests that identifying individuals who experience preseason anxiety and connecting them to support to reduce symptoms may also help reduce rates of athletic injury.[30]

When preparticipation physicals are conducted, brief screening measures can be used to assess for potential mental health concerns and/or measure well-being to enable referrals to licensed mental health providers. The NCAA Mental Health Best Practices document encourages the use of preparticipation screening measures to assess alcohol and cannabis use, depressive symptoms, anxiety, disordered eating behavior/eating disorders, sleep, and attention-deficit/hyperactivity disorder among student-athletes.[31] Upon experiencing an athletic injury, additional assessment of mental health and well-being symptoms can be undertaken and compared to these baseline values to help inform treatment decisions. If screenings are conducted, it is critical to have a plan in place to protect confidentiality, to ensure timely follow-up, and to know when and how to refer to a licensed mental health provider for the athlete's care (see Authors' Preferred Technique for more information).

DECISION-MAKING PRINCIPLES

For all front-line sports injury professionals, it is important to recognize potential mental health concerns. Rather than treating these issues as taboo, open discussion among all providers in the training room setting can validate the importance of psychological issues, the active role each provider can take in directly intervening, and the importance of referring to other resources when available. Taking an active role in supporting athletes directly makes a difference in their recovery. Professionals can also address pressures to return to sport (direct or indirect) from coaches, parents, teammates, administration, and athletes themselves to shape the influence of these assets and to enhance a healthy recovery process.

Caring for the injured athlete ideally involves collaboration between many potential resources. These include the athlete him/herself, sports physicians, Certified Athletic Trainers (ATCs), sports nutritionists, sports psychologists and/or licensed mental health providers, administration, coaches, teammates, family members, professors, academic coordinators, and other potential support systems. Integrating these potential resources into a multi-disciplinary team with well-defined roles and recognition of role limitations can be invaluable in addressing the complete biopsychosocial needs of the recovering athlete.

It is important to note that only licensed mental health providers are qualified to focus on mental health concerns while a qualified sport psychology consultant can focus on skill-building for sports performance while the athlete is returning from athletic injury. A qualified licensed mental health provider with training and experience working with athletes can fluidly work on both mental health concerns and skill-building for performance during the return-to-sport process. While it is ideal that all injured athletes have access to properly trained and credentialed professionals serving these roles, such resources do not exist at all levels across the breadth of athletics. However, consultations and/or referrals to these providers through access to school resources not specific to the athletics department and/or in the surrounding community can also be a great fit.

There are a number of potential challenges that can prevent athletes from seeking care with a mental health provider, such as the role of stigma or the belief that seeking care would be a sign of "weakness," not wanting to express emotions, lack of time to seek care, being unsure how to access a provider who is competent in working with athletes, and/or the belief that treatment may not help.[32] The role of sports medicine personnel is critical in helping athletes overcome these barriers. When referring to a mental health provider, it is often helpful to meet with an athlete individually, to use "I" statements, to focus on observed behaviors/events, to ask the athlete how they are doing and listen before moving to problem-solving, to emphasize the athlete's value to the team despite the athletic injury, to be clear about how to contact the mental health provider and what the treatment may look like, to talk about the confidentiality of available mental health services, and to follow-up after making the referral to see if the athlete was able to get connected to a mental health provider.

Authors' Preferred Technique

Treatment Options

A number of options exist for supporting athletes' psychological needs in their return-to-sport process, including techniques that focus on building skills for sport performance, strategies that focus on mental health concerns, and approaches that address both concerns. Techniques that focus on sport performance when an athlete is recovering from athletic injury often include goal-setting, the use of imagery or visualization exercises, teaching the athlete about self-talk, improving confidence in current and future athletic capabilities, and reframing the injury as an opportunity to learn and develop a new relationship with sport that can be positive. In a study conducted with 1283 athletes from the United States, the United Kingdom, and Finland, only 27% of athletes ($n = 346$) reported using mental skills during injury rehabilitation, but of those who used these skills, 72% ($n = 249$) said they believed the mental skills helped them rehabilitate faster.[33] The top four skills these athletes reported using included goal setting (46.8%), positive self-talk (33.2%), imagery (31.8%), and relaxation (24.3%).[33]

A positive relationship has been found between goal setting and adherence to rehabilitation exercises during the recovery process.[34] When setting goals, use of the SMARTS principle[35] is a pragmatic approach that can be used by many different members of the athlete's multi-disciplinary team. An injured athlete's attention is directed to *S*pecific, *M*easurable, *A*ction-oriented, *R*ealistic, *T*ime-based, and *S*elf-determined indicators of progress to enhance perceptions of competence and facilitate a healthy and efficient return to sport process. When goals do not meet the SMARTS criteria, the normal anxiety and fear experienced by the injured athlete may undermine adherence to treatment and lengthen the rehabilitation process. Again, having an established consultation relationship with a licensed mental health provider can be invaluable when an athlete experiences significant changes in anxiety or depression over the course of recovery.

Techniques that focus on mental health concerns or both mental health and athletic performance concerns at the same time often target anxiety symptoms (including pre-performance anxiety, fear of reinjury, and worry about not being able to perform at preinjury levels); depressive symptoms and feelings of loss; feelings of isolation, anger, relationship conflicts that emerged as a result of the injury and/or removal from sport participation; changes in eating behavior; changes in sleep; addressing problematic coping strategies such as substance abuse; and fostering the use of positive coping strategies. Exploration of identity outside of sport also can be critical for the athlete's well-being, both for those who are temporarily removed from athletic participation and for those whose have suffered career-ending athletic injuries. Podlog et al.[28] highlight the importance of promoting a perspective shift from active sport participation as the sole basis for team membership. Affiliation can be redefined through work with a licensed mental health provider in the context of reexamining the intrinsic value of sport participation, redefining competence as a contributing teammate, exploring the notion that active sport participation is only one facet of the athlete's identity (i.e., what else is part of who you are?), and supporting an athlete in identifying "transferrable skills" from sport to other elements of their identity. In some instances, this can also be supported through actions by sports medicine staff, coaches, and teammates, such as creating a new interim or permanent role on the team (e.g., volunteer coach, training partner) and through building relationships with other injured athletes through a formal venue (e.g., a support group) or through informal exchanges with others who have been through this experience before.[28]

While the strategies focusing on mental health concerns should only be practiced by an appropriately trained and licensed mental health provider, having all treatment team members know how to speak to an injured athlete about options for care and about what treatment may entail can help to ensure a successful referral and allow holistic support to be provided to the athlete throughout the return-to sport process. There are a number of different theoretical approaches that mental health providers may use, which can be beneficial for injured athletes, including cognitive-behavioral therapy (CBT), interpersonal therapy (IPT),[36] positive psychology, mindfulness-based stress reduction (MBSR), acceptance and commitment therapy (ACT), humanistic therapy, and more. For example, a cognitive-behavioral therapist may work with an athlete on ways to identify automatic negative thoughts that emerge related to athletic injury and the return-to-sport process, and teach the athlete how to use thought-challenging techniques to determine an alternative thought that can be associated with a different and more desired emotional response.[37] Stress inoculation training has also been proposed as an intervention using cognitive restructuring, thought control, imagery, and simulation.[38] Athletes who were randomly assigned to a similar intervention, Cognitive Behavioral Stress Management, who received education on CBT techniques and relaxation strategies had significantly fewer illness and injury days compared to the control group of athletes. Therefore the proactive use of these strategies may also reduce the risk for injury and/or teach athletes a set of skills they can build upon should they experience athletic injury in the future.[39]

A therapist using acceptance and commitment therapy may focus on the goal of accepting both positive and negative thoughts and emotions without needing to change them, and may also use mindfulness techniques.[40] Positive psychology work can be integrated by focusing on the athlete's strengths and how these can be helpful during the recovery process.[41] Many athletes excel at being goal-oriented, open to feedback/coaching, maintaining motivation, being able to push through physical pain to enhance athletic performance, and managing time efficiently. Each of these skills can be invaluable during the recovery process, but may need to be adapted in new ways that are less familiar to the athlete. Treatment team providers who remind athletes of their strengths will help recovering athletes direct their attention from the deficits resulting from the athletic injury and removal from sport participation to the opportunities to facilitate their own recovery process and to minimize challenges.

In addition, interventions that modify physiological arousal following injury are thought to reduce the attentional disruption associated with a stress response to injury and improve focus. These benefits have been explored in using progressive muscle relaxation (PMR), meditation, autogenic training, and diaphragmatic breathing exercises.[42] They have been shown to enhance concentration and decrease distractibility during the recovery process.[43]

It has also been proposed that improving social support, teaching coping skills by training injured athletes and coaches, and building confidence will increase resilience to athletic injury.[44] Evidence is growing in other related recovery areas that argues for the benefit of combining biological and psychosocial approaches—areas as diverse as behavioral research in oncology[45] as well as premorbid[46] and comorbid[47] psychological contributions to mild traumatic brain injury. Despite the existence of social support, which shows great value in the psychological recovery from sports injury,[17,48,49] it may not be perceived as available.[50] Ensuring that education and information are not only available, but also perceived as available is key in managing sports injury recovery. In addition, different types of injuries may be associated with different levels of satisfaction with social support. While athletes who experienced orthopedic injuries and concussion injuries reported similar sources of social support, those who experienced orthopedic injury expressed more satisfaction with their support than the group who experienced concussion.[49] It may be that some of these social support systems lack knowledge of typical symptoms of concussion or are unsure of how to support someone experiencing concussion symptoms. The invisibility of concussion injuries also may allow social support systems to "forget" that the need for support exists, or can result in greater stigma for an athlete who is removed from sport participation without a visible wound, brace, or crutches. Sports medicine personnel can offer social support by keeping the injured athlete involved in the team by including supportive medical providers, friends, or family members in the rehabilitation process and through collaborative goal-setting with the athlete throughout the rehabilitation process.[51] In addition, providing frequent communication and education about why a particular rehabilitation process is used supports injured athlete autonomy. Updates about measured progress (SMARTS goals) and offering an injured athlete regular choices about the rehabilitation plan can reduce pressure and enhance the perception of an internal locus of control.

POSTINJURY MANAGEMENT AND OUTCOME

Continuing to support injured athletes after their initial return to sport is important in providing them with coping strategies for their worry about experiencing another athletic injury and their concerns about their current and future level of play. Athletes who have experienced an injury that required more time away from sport, and/or repeated athletic injuries, often need additional support. Athletes who have experienced a severe injury express more fear of returning to sport and more fear of reinjury.[52] Athletes who experienced anxiety about potential reinjury more frequently and more strongly were also more likely to decrease the time and effort they put into their rehabilitation exercises and treatment plan.[53] This can then increase their likelihood of experiencing reinjury and/or add additional stressors to their return to sport experience (e.g., limited range of motion, decreased flexibility, and/or lower athletic performance). Therefore there is a need for continued collaboration and involvement between a qualified sport psychologist or licensed mental health provider and the athlete after his/her physical return to sport.

When athletes were asked to define what a "successful" return to sport meant to them, answers included a number of performance measurements (e.g., a return to preinjury athletic performance and having realistic athletic goals postinjury), psychological measurements (e.g., "staying on the right path" of believing that one could still achieve long-term success postinjury, and feeling the "self-satisfaction" of meeting their own internal standards of success), and physical measurements (e.g., remaining uninjured and feeling confident that the injured body part is fully recovered).[26] A focus group conducted with injured athletes reported three aspects of psychological readiness to return to sport: feeling confident returning to sport, having realistic expectations about their current athletic capabilities, and being motivated to achieve performance standards that they had previously obtained prior to athletic injury.[54] These findings also highlight the important roles that sports medicine staff have in continuing to support the athlete after the initial RTP, as these athletes reported that trust in their rehabilitation providers, the achievement of physical outcomes/measurement points, goal setting, and reducing boredom during the rehabilitation process were important contributors to these three aspects of readiness to return to sport.[54]

COMPLICATIONS

Despite the efforts of athletes, coaches, and treatment team members to reduce complications throughout the return-to-sport process, real or imagined time pressure can influence the psychological and physical response to injury.[28] Athletes may perceive direct pressure from coaches, teammates, and/or family members; may perceive indirect pressure on their coaches from others; and/or may project their own desire for a rapid RTP onto their coaches. This places athletes in a vulnerable position to rush their RTP timeline ineffectively and dangerously. Coaches themselves may directly or indirectly convey perceived time pressure (from administrations, team owners, parents, boosters, or even the press) to have their athletes ready for the next competition. These pressures can exacerbate an athlete's fears of losing their abilities, "losing ground" relative to others who are not injured, losing a spot on the team, and losing "right of membership" on the team. Pressures, expectations, and cognitions regarding the past (i.e., what my abilities were and who I was) and the future (i.e., who I was on track to be, who I am supposed to be, who I "have" to be) lead to maladaptive arousal that detracts from the body's performance of recovery in the present moment. For the injured athlete, rest, recovery, refueling, and rehabilitation often take precedence over testing their new limits of strength and fitness. If an injured athlete is externally guided or internally compelled to push beyond the body's capabilities in the present, recovery is jeopardized. These pressures may push the athlete to rush the return-to-sport timeline, and paradoxically extend the return-to-athletics participation process[55] as the athlete may experience reinjury, illness, and/or perform poorly due to an incomplete recovery that continues to limit physical and psychological performance. Reducing return-to-sport pressures by talking openly to injured athletes, coaches, and family members about the expected return-to-sport timeline—and the reasons for this timeline—will allow everyone to be on the same page about the anticipated return and the expectations of the return-to-sport process.

FUTURE CONSIDERATIONS

Medical providers are often the staff members who interact with injured athletes earliest after their experience of athletic injury and most frequently throughout the rehabilitation and return-to-sport process. Therefore the sports medicine team (consisting of ATCs, team physicians, specialists, and other medical providers) is in a key position for establishing a multidisciplinary team that can holistically support the injured athlete during and beyond his or her return-to-sport process. Consulting with and referring to other providers as needed throughout the process, including a licensed mental health provider or qualified sport psychologist, can help reduce the negative impacts of athletic injury, and to facilitate a successful and timely return to sport. It is important to identify the gaps in your care model that leave unmet needs for injured athletes so you can determine appropriate solutions.

While this is not a new concept, and despite increased sport participation across all competence levels, there are too few resources aimed at easing the psychological impact of sports-related injuries. We hope that chapters such as this increase awareness of the importance of this issue, and encourage all those working with athletes at all levels to incorporate such conceptual understanding into their practice.

For a complete list of references, go to ExpertConsult.com.

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Citation:
Singer RN, Hausenblas HA, Janelle CM, eds. *Handbook of Sport Psychology*. Wiley; New York: 2000.

Level of Evidence:
I/II

Summary:

Chapter 30 Psychology of Injury Risk and Prevention (Jean M. Williams). Revised version of the stress and injury model (Williams and Anderson, 1998).

Chapter 31 Psychology of Sport Injury Rehabilitation (Britton W. Brewer). Presents the biopsychosocial model of sport injury rehabilitation (Brewer, Andersen, and Van Raalte).

Citation:

Weiss MR. Psychological aspects of sport-injury rehabilitation: a developmental perspective. *J Athl Train*. 2003;38:172–175.

Level of Evidence:

III

Summary:

Brief review of developmental factors in sport injury. Provides a brief background of developmental perspective, as well as an overview of research design, sampling procedures, and measurement/statistical issues in sport injury rehab. Recommends more research in this area, especially given the widespread nature of this type of injury.

Citation:

Yang J, Peek-Asa C, Lowe JB, et al. Social support patterns of collegiate athletes before and after injury. *J Athl Train*. 2010;45:372–379.

Level of Evidence:

I/II

Summary:

Prospective study that examined pre/post injury social support among male/female collegiate athletes.

Citation:

Podlog L, Dimmock J, Miller J. A review of return to sport concerns following injury rehabilitation: practitioner strategies for enhancing recovery outcomes. *Phys Ther Sport*. 2011;12:36–42.

Level of Evidence:

II

Summary:

Review of psychosocial stresses in returning athletes. Their review suggests that athletes returning to sports from injury may have difficulty with sense of competence, autonomy, and relatedness (Self-Determination Theory).

Citation:

Putukian M. The psychological response to injury in student athletes: a narrative review with a focus on mental health. *Br J Sports Med*. 2016;50:145–148.

Level of Evidence:

III

Summary:

Reviews trends in mental health concerns postinjury and barriers for student-athletes in seeking care, plus the importance of educating medical staff to recognize mental health concerns, including through preparticipation screening, and referring to mental health resources.

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